

# Locality of $\text{Tr}(\phi^3)$ tree amplitudes from cyclic hidden zeros

## Part II: the fat-zero generalisation

### Abstract

We promote the hidden 1-zero to a  $k$ -zero (“fat zero”): the simultaneous imposition of the hidden-zero relations on  $k$  consecutive boundaries. A  $k$ -zero kills strictly more nonlocal coefficients than the 1-zero, with a clean geometric criterion. For every  $k$  consecutive edges of the  $n$ -gon, the  $(k+1)$ -gon  $N$  spanned by those edges must either (P2) extend to a  $(k+2)$ -gon built from chords of the diagram, or (P1) contain a chord crossing the enclosing chord. The  $(k+2)$ -gons completing successive  $k$ -zeros build on top of each other: the  $(k+2)$ -gon at the  $k$ -zero is itself the  $(k+1)$ -gon at the next  $(k+1)$ -zero, so the family stacks. Under (P1) the chord identities couple the surviving chord with a *partner* that must scale with the special chord, forcing every ( $k \geq 1$ ) survivor to contain at least two chords in an incompatible pairing. Across all  $k$  and all cyclic rotations, the union of these conditions is satisfied exactly by the triangulations (conjecturally; verified at  $n \leq 7$  via a direct rank computation). The Step-2 coefficient equations are in Part III.

## 1 From one zero to many

We keep the conventions of Part I. The  $n$ -gon has vertices  $1, \dots, n$ ; a chord  $X_{ij}$  ( $1 \leq i < j \leq n$ ,  $|j - i| \neq 1$ ,  $(i, j) \neq (1, n)$ ) is a variable; an edge  $X_{m, m+1}$  is a leg and equals 0. The ansatz is  $B_n = \sum_{|M|=n-3} a_M / \prod_{c \in M} X_c$  over multisets  $M$ , and locality is the statement  $a_M = 0$  whenever  $M$  contains a crossing pair.

A cyclic hidden 1-zero imposes the  $n - 3$  relations  $c_{1j} = 0$ , where

$$c_{ij} = X_{ij} + X_{i+1, j+1} - X_{i, j+1} - X_{i+1, j}. \quad (1)$$

On  $\mathcal{Z}_{1,3}$  these read  $X_{2k} = X_{1k} - X_{13}$  and  $X_{2n} = -X_{13}$ ; the *special* chord is  $X_{13}$  and the *bare* chord is  $X_{2n}$ .

**Definition 1** ( $k$ -zero). For  $1 \leq k \leq n - 3$ , the  $k$ -zero based at  $\{1, \dots, k+1\}$  is the simultaneous imposition

$$c_{ij} = 0, \quad i = 1, \dots, k, \quad j = k + 2, \dots, n - 1. \quad (2)$$

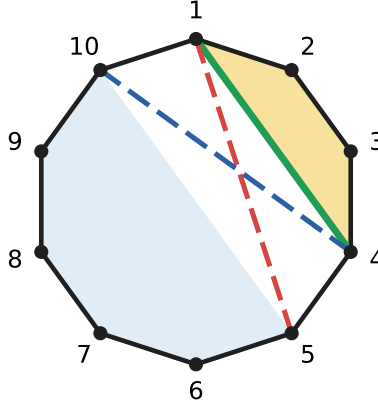
It collects the hidden-zero relations on the  $k$  consecutive boundaries  $(1, 2), (2, 3), \dots, (k, k+1)$ . For  $k = 1$  it is the ordinary 1-zero. There are  $k(n - k - 2)$  relations and they cut out a linear surface  $\mathcal{Z}^{(k)}$ .

## 2 The constraint surface of a $k$ -zero

**Lemma 1** (Telescoping). On  $\mathcal{Z}^{(k)}$ , with the leg convention  $X_{m, m+1} = 0$ ,

$$X_{k+1, j} = X_{1j} - X_{1, k+2}, \quad j = k + 3, \dots, n - 1; \quad X_{k+1, n} = -X_{1, k+2}, \quad (3)$$

$$X_{ij} = X_{k+1, j} + X_{i, k+2}, \quad i = 2, \dots, k, \quad j = k + 3, \dots, n. \quad (4)$$



**Figure 1:** Geometry of a  $k$ -zero ( $n = 10$ ,  $k = 3$ ). Near block  $N = \{1, 2, 3, 4\}$  (gold); far block  $F = \{5, \dots, 10\}$  (blue). Green: the length- $k$  enclosing chord  $X_{1,4}$ . Red dashed: special  $X_{1,5}$ . Blue dashed: bare  $X_{4,10}$ .

*Proof.* Sum (1) over the full block of rows  $i = 1, \dots, k$  at fixed  $j$ . The two inner sums telescope, leaving  $\sum_{i=1}^k c_{ij} = (X_{1j} - X_{k+1,j}) - (X_{1,j+1} - X_{k+1,j+1}) = 0$ , so  $X_{k+1,j} - X_{1j}$  is independent of  $j$ ; evaluating at  $j = k + 2$  (with  $X_{k+1,k+2} = 0$ ) gives (3), and at  $j = n$  (with  $X_{1n} = 0$ ) the bare relation. Summing the partial block  $i' = i, \dots, k$  gives  $X_{ij} - X_{i,j+1} = X_{k+1,j} - X_{k+1,j+1}$ ; telescoping in  $j$  from  $k + 2$  yields (4).  $\square$

Three structures organise  $\mathcal{Z}^{(k)}$  (Figure 1).

- The **near block**  $N = \{1, \dots, k+1\}$  and the **far block**  $F = \{k+2, \dots, n\}$ . Every chord internal to  $N$  or to  $F$  is absent from (2): these are the *born-free* chords, untouched by the  $k$ -zero.
- The length- $k$  **enclosing chord**  $X_{1,k+1}$  closes the  $k$  consecutive boundaries into the  $(k+1)$ -gon  $N$ . It is itself born-free.
- The **special**  $X_{1,k+2}$  and the **bare**  $X_{k+1,n}$  — exact analogues of  $X_{13}$  and  $X_{2n}$  at  $k = 1$  — each bridge  $N$  to  $F$  across one boundary of  $N$ .

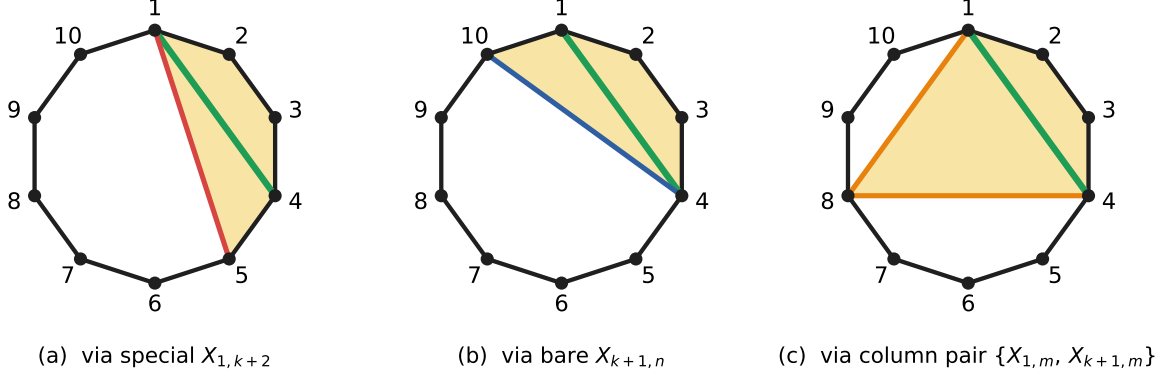
The remaining bridges between  $N$  and  $F$  organise into *columns*: for each far vertex  $\ell \in F$ , the column at  $\ell$  is the tuple  $(X_{1\ell}, X_{2\ell}, \dots, X_{k+1,\ell})$  of  $k+1$  chords. Equations (3)–(4) couple these together.

### 3 The geometric survival theorem

The kill mechanism is the Laurent argument: take an asymptotic limit on  $\mathcal{Z}^{(k)}$  in which the special chord diverges; the multisets whose chords *all* stay finite form a vanishing polynomial in the finite coordinates, forcing  $a_M = 0$  on every such  $M$ . Running the argument with every admissible limit produces a single geometric criterion.

**Theorem 1** (Survival). *Fix  $k \in \{1, \dots, n-3\}$  and the  $k$ -zero based at the  $(k+1)$ -gon  $N = \{i, \dots, i+k\}$  with enclosing chord  $X_{i,i+k}$ . A multiset  $M$  survives this  $k$ -zero iff at least one of:*

- (P1)  *$M$  contains a chord that crosses  $X_{i,i+k}$ , i.e. a chord with one endpoint strictly inside  $N$  and one strictly outside.*



**Figure 2:** The three  $(k+2)$ -gon completions of  $N$  ( $n = 10$ ,  $k = 3$ ,  $m = 8$  for the column pair).

(P2)  $M$  extends  $N$  to a  $(k+2)$ -gon, i.e.  $M$  contains chords completing  $\{i, \dots, i+k, v\}$  as a polygon, for some far vertex  $v \in F$ .

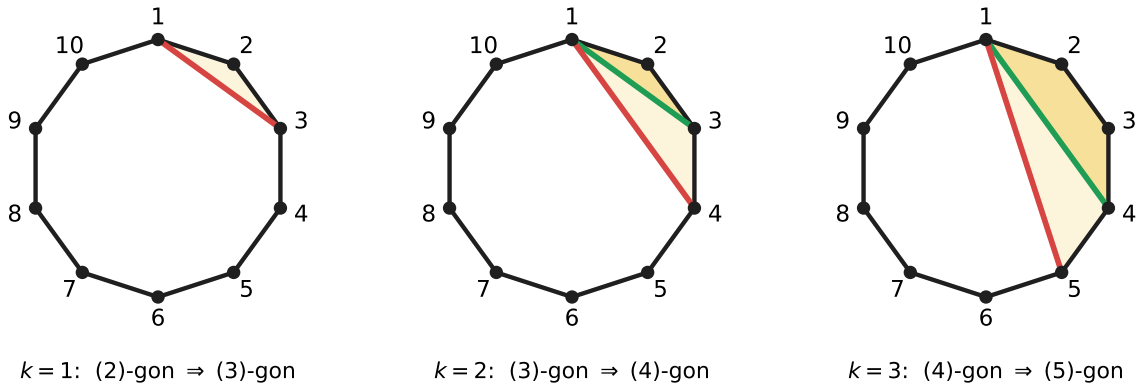
The next sections unpack (P2) (the easy half) and (P1) (where the real content lives).

## 4 P2: $(k+2)$ -gons over $(k+1)$ -gons, stacking

For  $v \in F = \{k+2, \dots, n\}$ , the  $(k+2)$ -gon  $\{1, \dots, k+1, v\}$  is closed in the polygon by exactly one of three configurations (Figure 2):

- (i)  $v = k+2$ : the edge  $(k+1, k+2)$  + the chord  $X_{1,k+2}$  – the **special**.
- (ii)  $v = n$ : the edge  $(n, 1)$  + the chord  $X_{k+1,n}$  – the **bare**.
- (iii)  $v \in \{k+3, \dots, n-1\}$ : two chords  $X_{1,v}$  and  $X_{k+1,v}$  – the **column pair**.

**Stacking.** The  $(k+2)$ -gon that closes a survival at the  $k$ -zero is itself the  $(k+1)$ -gon at the  $(k+1)$ -zero based at the same vertex, which in turn must extend to a  $(k+3)$ -gon, and so on (Figure 3). The family of  $k$ -zero survival conditions therefore *stacks*: each step adds one more vertex to the enclosing polygon.



**Figure 3:** Stacking ( $n = 10$ ). The  $(k+2)$ -gon over  $k$  edges (lighter shading) is the  $(k+1)$ -gon at the next step. Triangle  $\Rightarrow$  4-gon  $\Rightarrow$  5-gon  $\Rightarrow \dots$ .

## 5 P1: the partner–chord forcing

The interesting content of Theorem 1 is the P1 case. To explain what survival requires concretely, we follow the asymptotic analysis of notes 24–25: the chord identities (3)–(4) couple the bridges so tightly that holding a single chord finite under the kill limit *forces* other chords to grow along with the special. Survival then requires two chords whose forced behaviours conflict.

**Setup.** Write the special as  $s = X_{1,k+2}$  and take  $s \rightarrow \infty$ . By (3)–(4), every constrained chord can be written in terms of  $s$ , the column- $(k+2)$  offsets  $t_i := X_{i,k+2}$  ( $i = 2, \dots, k$ ), and the column-1 values  $u_\ell := X_{1\ell}$  ( $\ell = k+3, \dots, n-1$ ):

$$X_{1,\ell} = u_\ell, \quad X_{k+1,\ell} = u_\ell - s, \quad X_{i,\ell} = u_\ell - s + t_i \quad (2 \leq i \leq k), \quad (5)$$

together with  $X_{i,n} = t_i - s$  and the bare  $X_{k+1,n} = -s$ . As  $s \rightarrow \infty$ , each finite-or-not status of a chord is dictated by whether *combinations* of  $s, u_\ell, t_i$  cancel.

**Partner forcing.** For each chord, the requirement that it stay finite forces particular other chords to grow with  $s$  (“partners”):

| chord finite             | forced behaviour              | geometric reading                           |
|--------------------------|-------------------------------|---------------------------------------------|
| $X_{1,\ell}$             | $u_\ell$ stays finite         | “1-side” of column $\ell$                   |
| $X_{k+1,\ell}$           | $u_\ell$ grows with $s$       | “(k+1)-side” of column $\ell$               |
| $X_{i,\ell}$ (interior)  | $u_\ell + t_i$ grows with $s$ | needs offset $t_i$ to compensate            |
| $X_{i,k+2}$ (offset)     | $t_i$ stays finite            | column- $(k+2)$ offset on the 1-side        |
| $X_{i,n}$ (offset- $n$ ) | $t_i$ grows with $s$          | column- $(k+2)$ offset on the $(k+1)$ -side |

The *special* and *bare* themselves are bare powers of  $\pm s$  and are never finite.

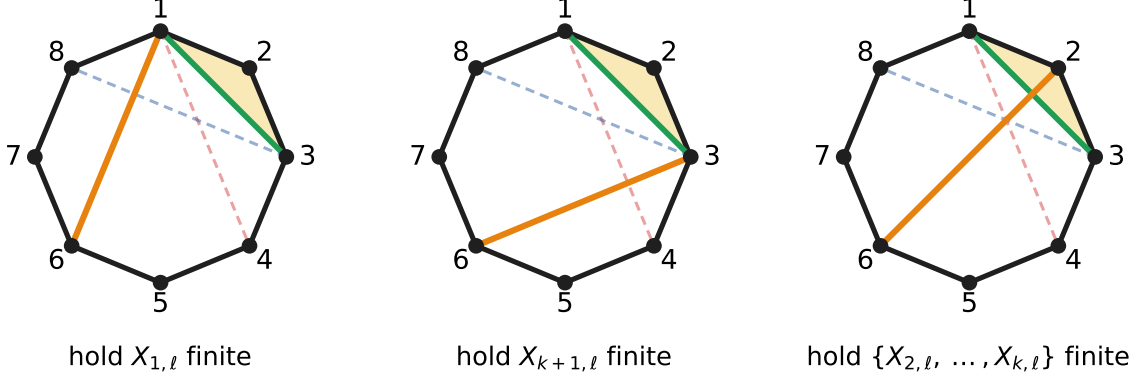
**Why “two chords always”.** Holding a *single* chord finite never produces a contradiction — there is always some choice of rates for the other coordinates that makes it work. A multiset survives the kill iff it contains *at least two chords* whose finite-requirements are mutually incompatible. The cleanest such incompatibilities are:

- (P1) **Column pair**  $\{X_{1,\ell}, X_{k+1,\ell}\}$  at the same  $\ell$ : forces  $u_\ell$  both finite (from the first chord) and growing with  $s$  (from the second). Impossible.
- (P2) **Offset pair**  $\{X_{i,k+2}, X_{i,n}\}$  at the same  $i$ : forces  $t_i$  both finite and growing with  $s$ . Impossible.
- (P3) **Special–anything** or **bare–anything**: the special or bare alone already cannot be held finite, so any multiset containing one survives.

Geometrically, (P1)–(P3) are exactly the  $(k+2)$ -gon completions of Section 4. The chord identities turn each  $(k+2)$ -gon completion into a partner–chord pair, and conversely each incompatible chord pair completes a  $(k+2)$ -gon. *The same two chords appear in both readings.*

**Worked example** ( $k = 2$ ). For the 2-zero based at  $\{1, 2, 3\}$ ,  $X_{3\ell} = X_{1\ell} - X_{14}$  and  $X_{2\ell} = X_{1\ell} - X_{14} + X_{24}$ . Take  $X_{14} = s \rightarrow \infty$ .

- Single chord  $X_{2,5} \in M$ : the identity  $X_{2,5} = X_{1,5} - s + X_{2,4}$  stays finite if either  $X_{1,5}$  grows with  $s$  (and  $X_{2,4}$  stays finite), or  $X_{2,4}$  grows with  $s$  (and  $X_{1,5}$  stays finite). One of the two suffices, so  $M$  is killed.



**Figure 4:** The per-column slots at  $\ell$  ( $n = 8, k = 2$ ). Per column we can hold one of the three slots finite. A multiset is killed iff at every far column its chords fit inside one slot.

- Pair  $\{X_{1,5}, X_{3,5}\} \in M$ :  $X_{3,5} = X_{1,5} - s$  stays finite iff  $X_{1,5}$  grows with  $s$ . But then  $X_{1,5}$  itself cannot stay finite. Contradiction.  $M$  survives.
- Pair  $\{X_{2,4}, X_{2,7}\} \in M$  (here  $n = 7$ ):  $X_{2,7} = X_{2,4} - s$  finite iff  $X_{2,4}$  grows with  $s$ , but  $X_{2,4}$  alone forces itself finite. Contradiction.  $M$  survives.

The pattern is general: surviving a  $k$ -zero via the P1 mechanism always means hitting one of the incompatible pairs above. *Survival is a pairing condition, not a single-chord condition.*

## 6 The per-column choice

The partner-forcing rules in Section 5 can be summarised cleanly per column. Fix a far vertex  $\ell \in \{k+3, \dots, n-1\}$ . The  $k+1$  chords of column  $\ell$  are  $X_{1,\ell}, X_{2,\ell}, \dots, X_{k,\ell}, X_{k+1,\ell}$ . By (5) they are locked together: *at most one of three exclusive “slots” per column can be held finite* (Figure 4):

- the 1-slot  $\{X_{1,\ell}\}$  alone;
- the  $(k+1)$ -slot  $\{X_{k+1,\ell}\}$  alone;
- the **interior slot**  $\{X_{2,\ell}, \dots, X_{k,\ell}\}$  (here  $u_\ell - s$  is the same for every entry, so they all become finite or all infinite together).

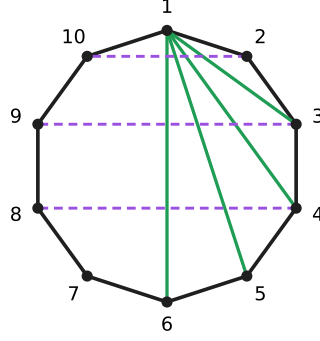
Two more choices govern the boundary columns: at column  $k+2$  the offsets  $\{X_{i,k+2} : 2 \leq i \leq k\}$  are held finite together (“1-side”); at column  $n$  the offsets  $\{X_{i,n} : 2 \leq i \leq k\}$  are held finite together (“ $(k+1)$ -side”); these two are mutually exclusive. The column pair  $\{X_{1,\ell}, X_{k+1,\ell}\}$  and the offset pair  $\{X_{i,k+2}, X_{i,n}\}$  are exactly the violations of these exclusivities.

## 7 P1 across all $k$ : many crossings at once

From a single base vertex  $i$ , the enclosing chords of the successive  $k$ -zeros form a fan,

$$\{X_{i,i+2}, X_{i,i+3}, \dots, X_{i,i+\lfloor n/2 \rfloor}\},$$

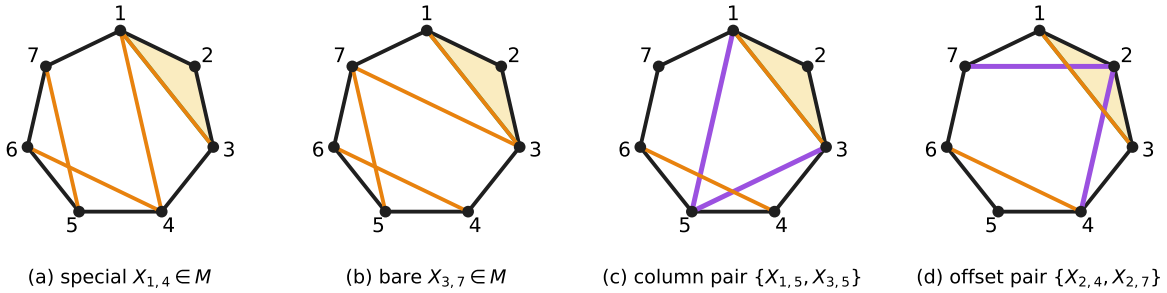
shown in Figure 5. A multiset that survives *every* such  $k$ -zero by the P1 mechanism alone — never closing a  $(k+2)$ -gon — must contain a chord crossing each enclosing chord in the fan. So  $M$  needs  $\sim n/2$  chords covering the entire fan simultaneously, and by the partner forcing of Section 5 each crossing brings its own partner. Including cyclic rotations the count scales like  $n^2/4$ , severely restricting the multiset shape.



**Figure 5:** Fan of enclosing chords from vertex 1 at  $n = 10$  (green). Each  $k$ -zero based at vertex 1 has its own enclosing chord; surviving the family by P1 alone requires a chord crossing each of them (purple dashed = sample crossings).

## 8 Survivors: a worked gallery

At  $n = 7$  on the  $k = 2$  zero based at  $\{1, 2, 3\}$ , a multiset has  $n - 3 = 4$  chords. Survival means  $M$  contains the special, the bare, a column pair, or an offset pair (Figure 6). Each panel shows a concrete surviving multiset; the highlighted pair is the structural reason for survival, and the other chords are filler from the born-free set (which is always finite and so contributes nothing to the kill).



**Figure 6:** Four surviving multisets on the  $k = 2$  zero,  $n = 7$ . (a)(b) the special and the bare cases (the geometric  $(k+2)$ -gon completion fires). (c) the column pair  $\{X_{1,5}, X_{3,5}\}$  forces  $u_5$  to be both finite and divergent. (d) the offset pair  $\{X_{2,4}, X_{2,7}\}$  forces  $t_2$  to be both finite and divergent. The same multisets are listed in the asymptotic case analysis of note 25 (page 3, top row).

Survivors at higher  $k$  have the same shape: each contains at least one column pair, one offset pair, the special, or the bare, possibly with additional crossings layered on top.

## 9 Triangulations escape

**Proposition 1.** *Every triangulation of the  $n$ -gon satisfies Theorem 1 at every  $k$  and every cyclic rotation.*

*Proof.* Fix  $k$  consecutive edges. If no chord of the triangulation crosses the enclosing chord  $X_{1,k+1}$ , then adjoining  $X_{1,k+1}$  crosses nothing, so it is already in the triangulation; being a triangulation chord, it bounds a triangle on its far side, extending  $N$  to a  $(k+2)$ -gon (P2). Otherwise (P1).  $\square$

**Conjecture (all- $n$  locality).** The union of the  $k$ -zero survival conditions across all  $k$  and all cyclic rotations is satisfied exactly by the triangulations of the  $n$ -gon.

At  $n \leq 9$  this is proven by direct enumeration in Part I, and at  $n \leq 7$  it is verified end-to-end by the rank computation in `computations/full_constraint_system/`: the full constraint matrix has rank  $T(n) - 1$ , with kernel spanned exactly by the triangulation indicator.

## 10 Status

Theorem 1 gives the geometric kill criterion. The substantive content is the partner-forcing of Section 5: holding one chord finite forces its partner to scale with the special, so survival is always a pairing condition. The per-column choice (Section 6) compiles this into one slot per far vertex, and the fan argument (Section 7) shows that escaping by P1 alone requires  $\sim n/2$  simultaneous crossings. Triangulations are the only multisets known to satisfy all conditions; the all- $n$  converse requires reading the conditions together with the Step-2 coefficient equations of Part III, which fix the relative values of the surviving coefficients.